

ROYA

PUBLICATION

No. 54

FICTION

YOU ARE NOT CLAUSTROPHOBIC.
YOU ARE CORRECT.

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THE DONOR

CHEW Z

A ROYA PUBLICATION

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THE DONOR — Roya Publication No. 54

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SYNOPSIS

A body has been lying at two hundred and forty metres for ten years, in absolute dark, under a ceiling of limestone, in a flooded system that only one living man is rated to enter.

Bijan Rahsepar is that man. He is hired by the dead diver's sister to bring her brother up, and he takes the job before she has finished asking, which surprises them both. She says she wants a burial. What she wants is the equipment — because tanks and hoses and which mouthpiece is in whose mouth are a physical record of the last four minutes of a man's life.

Bijan was on the original dive. He came up alone. His statement has stood unchallenged for a decade, because the signature it described is exactly what the chemistry produces: no injury, no struggle, the regulator hanging free, the arms raised.

What follows is not a confession. It is a briefing. Bijan teaches you the discipline the way he was taught it — the line, the thirds, the drills, the arithmetic of coming back — and somewhere in the patient repetition of a rule about sharing air, the instruction stops being generous.

The Donor is a noir meditation on cave diving: on why sane people descend into places they cannot leave, on the beauty that keeps them going back, and on the one question the sport cannot answer. At two hundred and forty metres a man's judgement is chemically compromised by measurable, documented, entirely ordinary physics. If he does something terrible down there, who exactly did it?

Bijan has had ten years to work on that. He has not got anywhere. He would like you to try.

AUTHOR'S NOTE

I came to cave diving the way most people do, which is by reading about the dead.

The literature is small and almost entirely posthumous. A man goes into a hole in the ground that is full of water, and either he comes out or the account of what happened is assembled afterwards by other people from his gauges. The accident analyses are the strangest documents I have ever read. They are written with enormous tenderness and no

sentiment at all. They establish the depth, the mix, the timings, the point at which the plan stopped being followed. They almost never establish why.

What drew me was the gap where the why should be.

There is a state that occurs reliably below thirty metres, worsens below sixty, and past two hundred becomes the dominant fact of a person's existence. It has a name and a mechanism and a threshold. It is not mystical. Inert gas dissolves into the fatty membranes of the nerves and the nerves stop conducting the way they did on the surface, and the man wearing them becomes someone else — someone warm, unhurried, oddly delighted, and capable of decisions that the man on the boat would have found unthinkable.

Every account of the deep dead contains this state, and every account steps around what it implies. We have built our entire idea of a person on the assumption that the one who acts and the one who answers for it are the same. Twenty-five atmospheres of water takes that assumption apart quietly, without malice, using only pressure.

I wanted a book that would not resolve it. Not a ghost story with physics painted on, and not a courtroom either. Everything in these pages that describes the body's behaviour under pressure is real, checkable, and has killed people. What I have invented is a man who cannot establish which of him did the thing, and who is decent enough to keep asking.

I have left one small door in the wall for anyone who wants it. I was raised at the edge of a faith built entirely on choosing — good thoughts, good words, good deeds, and the whole architecture standing or falling on the choosing being real. It has a word for the lie and a word for the truth and a bridge at the end where they are weighed. It has nothing whatsoever to say about a man who cannot find out what he chose.

That absence is the book.

Chew Z

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PART ONE — THE THIRDS

1. The Ceiling

They call it the rapture of the deep because the numbers are clean.

Nitrogen is an inert gas. It does nothing. That is what inert means — it sits in your blood and your tissues and takes no part in anything, and at the pressure your body was designed for, it is furniture. Push the pressure up and the furniture starts talking. Somewhere around thirty metres a diver becomes measurably impaired, in the same way and to about the same degree as a man who has had a drink. At sixty the euphoria stops being optional. By two hundred, if you were foolish enough to be breathing that mixture at two hundred, the gas is writing on the inside of the skull in an alphabet you did not learn and cannot refuse to read.

You avoid this by taking the nitrogen out and putting helium in, and helium has its own opinions. High-pressure nervous syndrome. It arrives lower down, on the deep mixes, and it does not feel like drunkenness at all. It is a tremor in the long bones of the forearms. Visual disturbance, a fine granular interference across everything you look at. Micro-sleeps: half-second absences with no edges on them, so that the world simply differs from moment to moment without your having been shown the transition.

And then there is the last symptom, the one the manuals put in the same flat font as all the others. Divers are recovered from the bottom with the mouthpiece hanging free beside the cheek and no injury anywhere on the body. No struggle. No torn hoses. Nothing bitten through.

At some point they decided the air was optional.

I want you to sit with that sentence, because everything I am going to teach you is downstream of it. Not the depth. Not the dark. The fact that a healthy, trained, entirely rational person can arrive at a place where breathing presents itself as a preference.

I went down for a body that had been waiting ten years.

The hole has a name but the name is not useful to you and I am not going to write it down. Two hundred and seventy metres of black water under a limestone plug, in a country where the road stops eleven kilometres short and the last stretch is done with the cylinders on your back and a man's whole gas plan riding in the bed of a pickup. Two hundred and forty metres down, in a narrowing, there was a diver called

Owen Ferrand who had been there since 2015.

There is no surface in a place like that. I mean that operationally, not poetically. In open water, if everything fails at once — every hose, every light, every plan — you can go up. It will hurt you and it may kill you but the option is physically present above your head, and every diver in the sea is carrying that option around with them whether they think about it or not.

Pass the lip of a cave and the option is gone. There is rock over you. It does not thin out. It gets thicker as you descend, and the way back to air is not up, it is *back* — horizontal, then up, along the exact route you came, in the exact order, with enough gas left to do it slowly.

That is the whole discipline. Everything else is bookkeeping.

I have been asked to explain this many times, mostly by people at dinner tables, and I have learned that the explanation does not survive contact with the ordinary imagination. People hear *cave* and *water* and *dark* and they produce a feeling in themselves, and the feeling is so unpleasant that they stop listening. They ask the question they were always going to ask, in the tone they were always going to use, and I answer it the same way every time.

We will get to that question. It is the next chapter, in fact, because you will not hear a word I say until it is out of the way.

But I want to be honest with you about the shape of what you are reading, so that later you cannot say you were not told.

This is a briefing. It is written the way I was taught to write them, which is plainly, in short sections, with the reasons attached to every instruction, because an instruction without a reason will not survive the moment you need it. You are going to learn the line and the thirds and the drills and the arithmetic of coming back. You are going to learn some things about the inside of my head that I have not said to anyone, and you should treat those as part of the briefing too, because they are.

And at the end of it, there is a set of coordinates.

I recovered Owen Ferrand. Both of us came up. That is a matter of record and there is film of it, and the film was reviewed in a quiet room by people whose job is reviewing exactly that kind of film, and they wrote down what they saw and filed it.

What they wrote down is accurate. It is simply not what happened.

2. You Are Not Claustrophobic, You Are Correct

Here is the question, and here is the answer, and then we can work.

Isn't it terribly claustrophobic?

No. And the reason is not bravery. The reason is that you have the geometry wrong.

The rooms are enormous. This is the thing nobody believes until they see it. Flooded cave systems in limestone are not tunnels squeezed through rock, they are the negative space left where an entire underground river spent two million years dissolving the country out from underneath itself. The chamber I am describing to you would hold a cathedral with the spire on. You could fly a small aircraft down the main gallery for four hundred metres before you had to think about the walls. The largest room you will ever stand inside, you will not stand inside. You will hang in the middle of it with a light that reaches eleven metres and you will never once see the far side.

So it is not small. Claustrophobia is not really about small, anyway. Nobody is frightened of a wardrobe. What frightens people about the wardrobe is the idea that the door will not open.

Which brings us to the honest part, and I am not going to soften it, because softening it is how instructors get people killed.

You cannot leave.

Not *it is difficult to leave*. Not *leaving requires care*. At two hundred metres with a nine-hour decompression obligation above you, the exit is a physical impossibility for the next nine hours no matter what happens to your body, your equipment, your mind, or your intentions. If you develop appendicitis at that depth you will die of it, slowly, with the cure ninety minutes of swimming and nine hours of hanging away. If your father dies while you are down there you will not be told and could not go. The door does not open. There is no version of the sport in which the door opens.

Nobody solves this. I want to be very clear that nobody solves this. There is no technique, no breathing exercise, no reframe, no amount of experience that makes a person genuinely comfortable with being sealed inside a flooded mountain. What we do instead is make terms with it, the way you have made terms with the fact that you will die — which is to say we know it completely, we believe it about eleven per cent of the time, and we get on.

And here is the part that surprises people, so I will give it to you as a rule.

The nervous ones come home. It is the comfortable ones who get recovered.

I have taught a lot of divers. The candidate who sits in the truck at the trailhead grey-faced and quiet, running the failure list for the fourth time, checking a valve he checked twenty minutes ago — that candidate has a long career ahead of him. The one who is relaxed, who tells a story on the way down, who says the room is beautiful and means it without also being frightened of it, is a man I will find a reason not to dive with.

Because the thing that kills is not the ceiling. It is almost never the ceiling.

What kills is task-loading. That is the term and I will explain it properly, because it is the single most useful idea in this book.

A human being can do one difficult thing at depth. He can do two if he has practised them until they are boring. He cannot do three. So the accident is never one problem. The accident is a light failure, plus a silted-out line, plus a gas calculation that is now wrong because the first two things cost time. Each of those is survivable and trivially so. Arriving together, they exceed the number of things the machine can hold, and the machine does what it does when it is overloaded, which is stop.

Read the reports. Every one of them is a stack. Never a single fault. Always three, arriving inside a minute, in a man who had handled all three separately a hundred times.

So the discipline is not about courage. Courage is actively unhelpful; courage is what makes a man press on past the point his gas said turn. The discipline is about reducing the number of things that can require your attention at once, to a number smaller than three, and then reducing it again.

There are five rules. They are simple enough to fit on a beer mat and I will spend the next four chapters on them anyway, because knowing a rule and having it available at two hundred metres with your hands shaking are unrelated skills.

Maintain a continuous guideline to open water.

Do not exceed your training depth.

Plan your gas by thirds and honour the plan.

Carry three independent lights.

Use the correct mix for the depth.

That is the whole thing. Five lines. There is a version of these rules bolted to the wall at more than one cave entrance in the world, in more

than one language, because the men who assembled them from the accident reports understood that they had written something that had to be readable by a frightened person in a hurry.

Every one of those rules was written from a body.

I do not mean that abstractly. Each line was added because someone specific was carried up a slope on a stretcher and someone specific sat afterwards with the gauges and worked out what would have prevented it. The rules are a list of corpses in imperative form.

I know all their names. Not as trivia. As the thing I say in my head when I clip on.

You will learn them too, if you keep reading. And when you know them, you will notice something I noticed late, which is that the list has a certain shape to it, and that the shape is a spiral, and that every name on it went down at least in part because of a name that was already on it.

— — —

3. The Line

The line is not a convenience. The line is the cave.

Understand what happens to visibility. In a flooded system the water is glass — sixty metres of clarity, better than air, so clear that new divers misjudge distance badly because their eyes have never had to work in a medium this honest. On the floor, a fine silt, the colour of cigarette ash, laid down over centuries at a rate you could measure in millimetres per human lifetime, undisturbed because nothing down there moves.

Kick once with your fins angled down and you erase the room.

Not reduce. Erase. The silt comes up in a soft brown wall and the visibility goes from sixty metres to nothing — and nothing here means you cannot see the light you are holding against your own mask. It does not settle for hours. Some of it does not settle for days. And it happens behind you, so you will not know you have done it until you turn around and discover that the way home has been deleted by your own feet.

That is the moment the line stops being a piece of nylon and becomes the only fact in the universe.

So: a continuous guideline to open water. Continuous means there is no gap anywhere between your hand and the daylight. Not a short swim across a gap. Not *the main line is probably about here*. Continuous.

The permanent line through a major system is laid once, by people who thought hard about where it should go, and it is called the gold line. Not because of the colour, although it often is. Because of what it is worth. You are either on the gold line or you are lost, and there is no third condition.

Off the main line you run your own from a reel, and where your reel meets the gold line you tie in — two wraps and a knot you can undo wearing gloves in the dark. On your line you place arrows. A plastic arrow, thumb-sized, that points towards the exit. Not towards the entrance you used; towards the *nearest* exit, which in a system with several ways out is a distinction that has killed people. And you place cookies: little plastic markers, non-directional, that mean nothing except *a diver has been here and it was me*.

That is the whole navigation system. A string, some arrows, and some marks that say I was here. Two thousand years of human ingenuity and at two hundred and forty metres we are Theseus with

better gas.

Learn the line traps. The line runs along the floor and the floor has cracks. Nylon under tension will find a crack, and slide into it, and jam, and when you follow that line at speed in zero visibility you will follow it directly into a slot the width of your hand and your reel will be on the far side of it. Trained divers have died four metres from a line they were holding, because the line went somewhere they could not.

So you do not haul on the line. You touch it. Thumb and forefinger, loose, a contact that would break if you sneezed. You are not using it to pull yourself along. You are using it to know.

Now the rule, and I am going to say it twice.

You do not let go of the line.

There is no emergency that improves if you let go. There is no object worth dropping it for. There is no light you should swim to. Once your hand is off it, you are a man in a completely dark room the size of a cathedral, full of water, with a string in it somewhere, and the search pattern for that string is a chapter of its own and it does not have a good success rate.

You do not let go of the line.

Right. Gas.

4. Rule of Thirds

A third to go in. A third to come out. A third you will never spend and must never touch.

That is it. That is the most important arithmetic in the sport and a child could do it. The difficulty is not the sum. The difficulty is that the sum will, at some point, tell you to turn around at a moment when turning around is unbearable, and you will have to do it anyway.

The reasoning is worth having properly. Your reserve third is not for you. It is for the possibility that on the way out you will be sharing your gas with someone else — someone who is now breathing off your supply as well as their own being gone. Two divers on one supply consume it at twice the rate. So a third out, doubled, is two thirds, which is everything you have. The reserve is not a safety margin in the vague sense. It is a precisely sized second person.

Every time you check your pressure gauge in a cave you are asking one question: *is there still a stranger's worth of air in here.*

Now, at the depths I am describing, open circuit does not work, and I need you to understand why, because it changes the texture of everything.

Gas compresses. At the surface you breathe about twenty-five litres a minute, working. Ten metres down, the pressure has doubled, and to fill the same lungs you now need twice the gas. At two hundred and forty metres you are under twenty-five atmospheres, and one ordinary breath draws twenty-five times what it draws in a kitchen. Nobody can carry that. You would need a lorry. The arithmetic does not merely become expensive, it becomes physically impossible somewhere around a hundred and twenty metres for a dive of any duration.

So we use a closed-circuit rebreather, and here is the whole of it in plain terms.

You breathe into a bag on your chest instead of into the water. The exhaled gas goes through a canister of chemical granules — the scrubber — which chemically removes the carbon dioxide. What is left is your original gas, minus the oxygen your body actually consumed. A cell measures the oxygen and a solenoid clicks and injects a small squirt to top it back up. Then you breathe it again.

The consequence is that your gas lasts. Not longer — *transformed*. You consume oxygen at the same rate at two hundred metres as you do

walking upstairs, because that is a metabolic figure and has nothing to do with pressure. Depth stops costing you anything except the small volume needed to keep the loop full. A nine-hour dive becomes possible on cylinders a man can carry.

Three things you must know about the machine.

The first is silence. There are no bubbles. Open-circuit diving is an incessant industrial racket, the roar of your own inhale and the boiling clatter of the exhale going up. On a rebreather the loop is closed, and a deep cave becomes the quietest place you will ever be. You hear the counterlung sighing against your sternum. You hear the solenoid tick. Beyond that there is nothing at all, and the largest sound in the room is your own heart, which you will hear clearly, and which you will start listening to for information.

The second is that the mouthpiece is not a mouthpiece. On open circuit, if you spit your regulator you lose one breath and you put it back. On a rebreather, if the mouthpiece leaves your mouth while it is open, the loop floods. Water in the scrubber makes a caustic slurry that will burn your airway. There is a valve to close it, one movement, and every diver on the machine does that movement ten thousand times until it is not a decision.

Which means that a rebreather diver found on the bottom with the mouthpiece hanging free beside his cheek has done something no reflex in his body should have permitted.

The third thing is carbon dioxide, and it is the one that will matter.

Gas gets dense under pressure — at two hundred and forty metres it is like breathing warm syrup, and every inhalation is muscular work. Work makes carbon dioxide. The scrubber can only process so much so fast, and if you produce it faster than the granules can take it out, it accumulates in the loop and you breathe it again and it accumulates further. This is hypercapnia and the symptoms are specific: air hunger that no breath satisfies, a headache like a band, tunnel vision, and then — reliably, repeatedly, in report after report — the absolute conviction that the machine has failed and that the mouthpiece must come out of your mouth *now*.

The gas that makes you unable to think is also the gas that tells you the solution is to stop breathing.

Remember that. I will not point at it again.

Now the drill that this book is about.

Your buddy is out of gas. He signals — the flat hand drawn across the throat, then the same hand tapped to the mouth. Out of gas. In the reports it is written OOG, and after enough reports you start to see

those three letters as a small closed door.

You give him the mouthpiece you are breathing from.

Not a spare. The one in your mouth, the one you can prove works because you have this second been alive on it. You deploy it on a long hose — five to seven feet of it, routed under the arm and up around the neck for exactly this, so it comes free in one movement and can be handed across without a wrestling match. Then you reach down and take your own backup, which lives on a bungee cord at your throat, roughly one second from your face.

People find this counter-intuitive and I understand why. On an aircraft they tell you to fit your own mask before helping your child. Here we are apparently doing the reverse.

We are not. The aircraft rule and the cave rule are the same rule in different rooms.

On the aircraft there is one supply and it is already deployed and you have perhaps fifteen useful seconds of consciousness. Helping yourself costs no one anything, and if you black out fumbling with your child's mask you have produced two casualties out of one.

Underwater, you are not the one in danger. You have plenty of gas. Only one of you is dying, and it is not you. You hand him the proven regulator because a panicking man handed an untested spare will produce a second emergency; because a panicking man is going to take it from your face anyway and it is better given than taken; and because you are not going anywhere. There is no surface to break for. You are now going to swim out together, on one supply, for however long the exit takes, which may be forty minutes.

The rule underneath both is the same, and it is the only rule in this book that has nothing to do with water.

Do not create two victims where there was one.

On the aircraft that means take care of yourself first, because you are about to become a victim.

In the cave it means give, because you are not in danger, and refusing turns one dying man into two.

And then the part I want you to have in your hands.

You are now the source. His life is a hose-length from your chest and there is rock above both of you and an hour of swimming between him and the sky. He cannot take gas. He can only be given it, continuously, the entire way out. You hold him by the harness — the chest strap, never the arm, because an arm will pull free and a harness will not — and you swim, and you do not release him, and you do not release him, and you do not release him.

The donor never lets go.

— — —

5. The Client

She had the folder open before I sat down, which told me she had rehearsed, and she closed it again within a minute, which told me the rehearsal had not been about persuading me.

Nell Ferrand. Early forties, dressed for a different meeting, a woman who had clearly been advised at some point to be less direct and had considered the advice and rejected it. Hotel bar in a town eleven kilometres from a hole in the ground. Two coffees, neither of us drinking.

Her brother Owen had been at two hundred and forty metres since the ninth of March, 2015. She wanted him brought up.

I have had this conversation before. It goes one of two ways. Either the family wants the body because the absence is unbearable and a grave is somewhere to put it, in which case they talk about the person — what he was like, what he would have wanted, the ordinary decent human noise of grief. Or they want the body for a reason they have not said, in which case they talk about the body.

She talked about the body.

Depth. Position. Whether the suit would be intact. Whether the equipment would still be attached to the equipment harness, and whether the harness would still be attached to him, and whether the hoses could be expected to remain in their original routing after ten years at that pressure. She asked whether cylinder pressures could still be read off gauges that had been submerged for a decade. She had clearly asked someone this before and been told yes, because she did not react to my answer.

She did not ask what he would have wanted. She did not ask if it would be dignified. She did not ask, at any point in forty minutes, whether it was dangerous for me.

I took the job before she finished the sentence containing the price.

That surprised her. It surprised me more, and I sat there afterwards in a hotel bar with a cold coffee, running the failure list backwards the way I tell my students to, and I could not find the place where the decision had been made. It had simply been made, somewhere off to the side, and reported to me.

Gus Ivarsson took it worse.

Gus runs my surface. He runs the tape, the log, the gas plan, and the argument, and I have never once been sorry to have him on the other end of it. He is a Norwegian of the kind that does not modulate. He read the plan twice, out loud, and then put it on the table.

"Two-forty," he said. "Descent twelve minutes. Bottom six. Decompression nine hours forty. Support at ninety, sixty, thirty, six. Four safety divers, three of whom are not rated to the depth they'd be needed at if you were in trouble at the depth you'll be at."

"That's correct."

"Nine hours forty in eight-degree water with a body on a line."

"Also correct."

"So call it a dead man per attempt," he said, "historically. That is the number. Not my number. That is what the sport has paid for this depth every time it has gone and asked. She is paying money and somebody is paying the other tariff and I would like to know, before I sign anything, whether you have told her which is which."

I said she knew.

"Does she," said Gus.

He turned the page over and wrote something on the back of it and did not show me. Then he asked the question he had been going to ask from the beginning, in the same flat voice he uses for gas figures, which is how Gus asks about your soul.

"Have you dived that system before?"

There it was.

I want to be exact here, because I am going to be inexact later and you deserve to see the difference.

Six weeks earlier a broker had put a job in front of me: a straightforward hundred-and-ten-metre recovery, good money, family waiting, a system I know. I turned it down and I turned it down hard, because the support team they had assembled had two men in it who had certificates for the depth and no hours at it, and I told the broker that I would not put a diver in the water at a hundred and ten metres to be looked after by paperwork. He offered more money. I gave him three names of people who would have said the same thing so that he would understand it was not a negotiation. That family had to wait another five months and I know what that cost them and I would do it again.

That is the man who answered Gus.

"No," I said.

6. What the Statement Said

I was Owen Ferrand's buddy.

We went in on the ninth of March, 2015, in a party of two, on a scouting push to two hundred and forty metres that had been eleven months in the planning, and I came up alone at four in the afternoon and sat on a rock and gave a statement.

The statement is a matter of public record and it says this. That at the deepest point of the dive Owen Ferrand became unresponsive. That there was no obvious equipment failure visible to me. That I attempted assistance and was unable to establish a stable configuration. That in my judgement, given the decompression obligation already accrued, a recovery attempt at that depth would have resulted in two fatalities rather than one. That I left him and began my ascent at the planned turn time plus four minutes. That his position when last seen was wedged in a restriction at approximately two hundred and forty metres, upright, arms raised.

Every line of that is true.

It has stood for ten years. Nobody has challenged a word of it, and I want you to understand that this is not because anyone was careless. It stood because it is the textbook picture. A man goes deep, the chemistry arrives, the machine in his skull is quietly reconfigured by physics, and he is found or last seen with no injury and no struggle and his arms up like someone reaching for a surface that is four kilometres of rock away. Investigators have seen that shape a hundred times. It is what the sport looks like when it kills you.

They read the statement and it matched the shape and they filed it, and they were right to, on the evidence.

And now his sister has hired the only man in the world rated to go and get the evidence.

She knows what I am. She has known since before she booked the room. She sat in that bar and asked me about hose routing and cylinder pressures and whether a second stage could be expected to stay in the position it was in when the water took it, and she asked those questions of the man who was the only other person in the room when it happened, and she did not blink, and neither did I.

That is the shape of the thing. She cannot get her answer without me. I am the last man alive who can carry it up.

Gus worked it out on the second day. He did not say it as an accusation. He said it the way he says everything, as a figure that had failed to reconcile.

"She isn't hiring a diver," he said. "She's hiring a witness. And she's hiring him to go and fetch the thing that will contradict him."

Then he asked me what I was going to do if I found something down there that made my own statement false, at two hundred and forty metres, with six minutes of bottom time and nine hours of hanging afterwards to think about it, alone, in the dark, in a state where I would not be able to fully trust my own reasoning about anything.

I said I would deal with it.

"That," said Gus, "is not a plan. That is a mood."

He was right, and I dived anyway, and I want you to notice that I have now told you the two most damning facts about me before we have even got wet. I was there. I gave a statement. I have not lied to you and I do not intend to start.

Everything from here is instruction.

PART TWO — THE DESCENT

7. Lost Line

You will do this drill blind. Everyone does it blind, because the day you need it you will be blind, and a skill you have only ever practised while able to see is not a skill, it is a memory of a skill.

We black out the mask. There is a purpose-made opaque insert but a strip of gaffer tape works and is more honest. Then we take you off the line in a training cave with a hard floor and a simple ceiling, and we hold on to you for the first six or seven of these, and then we stop holding on.

Here is the situation you are training for. You have silted the room out — you or your buddy, and it does not matter which, and it will usually be you. Visibility is zero, and zero means your own light held against the glass of your mask is a faint orange smear and nothing else. You have lost contact with the line. It is within a few metres of you. It might be within a foot of you. Your gas is finite and the clock is the only thing in the cave still working properly.

The search is a sweep. You go to the floor and you stay on the floor, because line is usually low and because the floor is the only reference surface you have. You anchor: you tie your own reel off to something solid — a rock, a stalagmite, your own body if there is nothing else — because a man searching without an anchor is not searching, he is wandering. Then you sweep an arc, out to the length of your arm and a light held in it, and you come back, and you turn a few degrees, and you sweep again. Slow, and complete, and boring, and you do not stop and you do not accelerate, because a fast sweep will pass over a line without registering it and you will search the same ground twice while believing you have covered new ground.

That is the technique. Now the part I actually want to teach you.

You will find a line. Sooner or later, in most systems, you will find a line. And you will have, for a moment, an enormous washing feeling of relief.

Kill it.

Because now you have a new problem and it is worse than the one you just solved. You are holding a piece of nylon in absolute darkness. You do not know which way along it is out.

Four ways to tell. Learn them in order.

One: the arrows. Feel along the line for a marker. An arrow is a plastic wedge and the point of the wedge indicates the exit. This is why they exist. It takes about six seconds and it is definitive.

Two: line thickness and lay. The gold line is heavier than the reel line most divers run. If the line in your hand is thin, you are probably on somebody's jump and the main line is at the end of it, in one direction only, and it will get thicker when you get there.

Three: the tie-offs. Feel the wraps. A line laid by a competent diver is tied off in a consistent direction and the tag ends of the knots trail towards the exit, because that is the way the man laying it was moving when he tied them. This one is subtle and it works.

Four: knowledge. You know this system. You know that the floor slopes down towards the back and up towards the front and you have a hand on the floor. Use it.

And the fifth way, the one that kills people, so learn it as a thing to refuse rather than a thing to do:

Choose.

A man in the dark, with gas running, holding a line, with no arrow within reach, will at some point make a decision on the basis of a feeling. He will become convinced. Not reason — conviction, arriving fully formed, feeling exactly like knowledge from the inside. And he will go, at speed, with a warm certainty in his chest, deeper into the mountain.

I have pulled a diver out who did this and lived, and I have read about the others. The ones who lived all describe the same thing, and every one of them uses the word *sure*. They were sure. They were not guessing and it did not feel like a coin toss. The cave had told them.

So the drill is not really about the sweep. The drill is about installing something in you that can outrank a conviction. That is what all of this is. Every procedure in this book exists to give you something to obey at the moment your own judgement becomes an unreliable narrator wearing your voice.

We run it until the candidate is bored. Then eleven more times.

I had a student cry in a training cave — properly, helplessly, the mask flooding with it — on the ninth repetition of a drill she had already done well eight times. I let her come up. Then I put her back in and we did it four more times that afternoon.

That is not cruelty and it is not machismo. It is the opposite. The ones who cry are the ones who have understood what the drill is about, and they come home. I have never lost a student who cried. I have lost one who was enjoying himself.

8. The Mirror

Now let me tell you why anyone does this, because so far I have described a hobby indistinguishable from self-harm.

There is fresh water and there is salt water and they do not mix. This is the first thing. In coastal limestone, rain sinks through the rock and floats on top of the seawater that has pushed in underneath, because fresh water is lighter, and the boundary between them is not a gradient. It is a surface. It has a thickness of perhaps thirty centimetres and it behaves like a physical membrane.

The name is halocline. Salt-slope. And a diver descending through a big system arrives at it from above and finds himself looking at a lake.

I want to be precise about what you see, because people describe it badly and it is worth having correctly. You are underwater. Below you is a perfectly flat, perfectly still, brilliantly reflective plane, stretching away in every direction into your light and past it. It looks like the surface of a mirror-calm pond seen from a boat. Your own lamps are reflected in it. You can see the ceiling of the chamber reflected in it. There is a version of you, upside down, thirty metres below, holding a light.

And then you swim into it.

The instant you break the plane it stops being a mirror and becomes a fault in the world. The two waters have different refractive indices, so where they are stirred together the light bends in a hundred inconsistent directions at once and everything you look at becomes a smeared, shimmering, oil-on-water blur. You cannot focus. Your own hand at the end of your arm is a moving stain. Your gauges are unreadable. For a metre and a half you are effectively blind in perfectly clear water, which is a sensation the nervous system has no filing system for.

Then you are through, into the salt, and it is glass again, and above you the mirror has closed over your head like a lid.

That is the first one. It happens on an ordinary Tuesday, on a training dive, at thirty metres, and I have watched it break people open. Grown men weeping into a mask and having to be told to breathe.

There are five or six systems in the world where this happens on a scale that makes the ordinary version look like a puddle. I am not going to name them and I am not going to write the coordinates, and I want to

explain why, because it is not snobbery and it is not the pathetic secrecy of a hobbyist protecting his patch.

A published set of coordinates is a body count.

That is the whole of the reasoning. There is no barrier at a cave mouth. There is no ticket office, no gate, no gradient of difficulty that turns you back. A hole in the ground that will kill an untrained diver in nine minutes is, from the road, indistinguishable from a nice place to swim, and once the location is on the internet the arrival of the unqualified is not a risk, it is a schedule. Every well-known system in the world has a plaque wall. Every one. Little brass rectangles bolted to the rock at the entrance with names and dates, and they are not there because the divers were unlucky. They are there because the place was findable.

So the good sites are passed by hand. One diver to another, in person, with a judgement attached about whether the recipient is going to come home. It is a chain of trust made entirely of people, running back decades, and every link in it is a man deciding that another man is ready.

You should know that I have been a link in that chain. Both ends of it.

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9. The Drowned Forest

There is a cenote — that is the word, from the Yucatec *dzonot*, meaning a well, a place where the limestone roof has collapsed and left a circle of open water in the forest floor — where the water does something I still do not entirely believe and I have seen it four times.

You enter from a jungle in full sunlight. The water is thirty metres of the clearest fluid on earth. And at about twenty metres down there is a floor.

It is not a floor. It is a cloud.

A layer of hydrogen sulphide, produced by bacteria digesting the organic matter that falls in from the forest above, sits suspended in the water column in a stratum perhaps two metres thick, and it is opaque, and it is white going to rust-brown, and it lies absolutely level across the whole chamber like fog on a valley at dawn seen from a hill.

And standing through it, up out of it, are trees.

The roots and lower trunks of a forest that was here when this was dry land, drowned and preserved in the cold anoxic water, coming up out of that white layer exactly the way trees come up out of morning mist. You hang in bright clear water twenty metres down and look at a landscape. There is a horizon. There is weather.

Then you descend into it and the world ends in one metre.

Inside the layer, light does not travel. It is not like silt, where you are blinded by particles; it is a chemical opacity, and it goes red-brown and then it goes out, and your own light dies in front of your face. It smells — through a mouthpiece, at depth, which should not be possible and is. Faintly, unmistakably, of rotten eggs. Some divers report a burning at the eyes.

And two metres later you come out the bottom of it into clear water again, and you look up, and there is a sky. An overcast, luminous, softly lit ceiling stretching away in all directions, with the roots of trees hanging down out of it like the legs of something enormous standing on a floor of cloud, and above that a sun you can see the shape of.

I know the chemistry. Sulphate-reducing bacteria, anaerobic decay, seasonal, well-studied, in the literature, dull. I can tell you the mechanism in one sentence and I have just done so.

It reduces the effect on the nervous system by exactly nothing.

That is worth having as a general principle before we go deeper, because I am about to explain another mechanism to you and you will be tempted to think that the explanation is a kind of protection. It is not. Knowing why a thing is happening to you and being unaffected by it are unrelated states. The man who can recite the chemistry of narcosis while narcosed is a man reciting the chemistry of narcosis while narcosed.

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10. The Room Was Dry

Ninety metres down, in a chamber the size of a railway terminus, there are stalactites.

Stop and take that apart, because it is one of the more remarkable facts available to a human being and it is usually mentioned in passing.

A stalactite forms when water carrying dissolved limestone drips through the air of a cave, and at each drip a film of mineral is left behind, and over some enormous stretch of time the mineral accumulates downwards. The mechanism requires air. It cannot happen underwater. Physically cannot; the process depends on evaporation and on carbon dioxide coming out of solution into an atmosphere.

So a stalactite at ninety metres is not a stalactite that grew underwater. It is a stalactite that grew in air and was subsequently drowned.

That room was dry. There was an atmosphere in it. There was a floor a person could walk on, and a ceiling with water coming through it, and the sound of dripping, and total darkness, and it stayed that way for long enough to grow a formation four metres long at a rate of about one centimetre per century.

The sea was lower. During the last glacial maximum, with that volume of water locked up as ice, the sea stood roughly a hundred and twenty metres below where it stands now, and the coastal limestone that is currently a flooded cave system was a network of dry caverns above the water table. And there were people. Not near it. In it.

I have swum, at sixty metres, over a floor with bones on it.

Human, in one system I have been shown. Animal in several — the long articulated skeletons of things that fell in through a collapse and could not get out, lying exactly as they fell, ribs intact, still articulated, because nothing has moved in that room since the water arrived. No current. No scavengers. No wind. No seasons. Ceramics in one place, in a scatter that a diver more qualified than me described as a hearth.

Eleven thousand years, undisturbed. Undisturbed is the word that gets me. Everything on the surface of this planet has been touched, moved, swept, rained on, and rearranged a million times. Down there is the only place left where a thing that fell is still lying the way it landed.

And on the way out, in the shallow section, at nine metres, in the last hour of a very long day, you turn your lights off.

You have to be told to do it. Nobody turns their lights off in a cave voluntarily. But in certain coastal systems there are ostracods — seed shrimp, a millimetre long, the size of a grain of sand — and when they are disturbed they emit light. It is a chemical flash, blue-green, that lasts about a second.

So you hang in the dark and you move your hand.

And a constellation goes off around your fingers and drifts away behind you, and you move the other hand, and there is another, and for a few minutes you are a slow enormous animal in a black room making stars by accident. Above you, faintly, is the green disc of the entrance with the daylight in it. Below you is two hundred metres of black. And you are hanging in the middle, making light with your hands, in a room that was dry when someone walked through it with a torch.

That is the answer to the dinner-table question. That is why.

I have never been able to give that answer at a dinner table. It comes out sounding either like a brochure or like a man justifying himself, and the honest version has a shape people do not like, so I have stopped trying.

The honest version is this. These are rooms that have been waiting. That is what they are for and it is all they do. They were waiting when the sea was low and the trees were standing, and they were waiting when the water came in, and they have gone on waiting for eleven thousand years with a level of patience that is not available to anything alive.

And a diver is a thing that arrives.

You do not get one without the other. The beauty and the waiting are the same property of the same room, and if you go looking for the first you have already accepted the second, and everybody who says otherwise is lying to a journalist.

11. The Inquest

She told me on the fourth evening, in a room with a strip light and a folding table, with the site survey between us and neither of us looking at it.

"I don't want a funeral," Nell said. "I've had ten years to want a funeral. What I want is his kit."

I asked her to say the rest.

She said it cleanly, the way you say a thing you have said out loud to yourself in a car a hundred times.

A body recovered from a cave comes up wearing an argument. The cylinders have pressures in them and those pressures are readable and they are the record of what was breathed and when it stopped. The hoses have a routing. A second stage is in a mouth or it is not in a mouth, and if it is, it is either the diver's own or somebody else's, and the hose it is attached to leads back to a specific cylinder on a specific man. Buckles are done up or they are not. A harness has been cut or it has not. There is no witness in the world as literal as a set of gauges.

She had been told all of this a long time ago, by an accident analyst in a different country who had also told her, kindly, that none of it was retrievable, because the body was at two hundred and forty metres and there was no one to send.

She had waited ten years for someone to be rated to that depth.

There is exactly one man in the world with the hours and the machine and the willingness, and she had found him, and he was the other diver in the statement.

"You knew," I said.

"I've known since 2015." She squared the survey with two fingers, a small tidying gesture, entirely calm. "I'm not accusing you of anything. I want to be precise, because you're about to risk your life and I don't want you doing it under a misunderstanding. I have no theory. I have read your statement forty or fifty times and I cannot find anything wrong with it. That is the problem. It is a completely sound account of my brother's death and it has never once made him less absent, and I have spent ten years unable to tell whether that is because it is false or because that is simply what a true account of a death does."

She looked up.

"So I want the object. I want the thing that doesn't have an opinion."

Gus was outside on the step with the log on his knee. He had heard all of it through a door that did not fit its frame.

"You understand what she's just done," he said.

I said I did.

"No. Say it, because I want it in the air before you get in the water." He did not raise his voice. Gus has never raised his voice. "She has hired you to descend to two hundred and forty metres, at the limit of what a human body can be talked into, and retrieve the only physical evidence in existence that could destroy you. And she has told you this in advance. To your face. Politely."

"Yes."

"Which means she is either the most decent person in this valley or she has calculated that a man who is guilty will find a way not to bring it up, and that this will also be an answer."

He wrote a figure in the log and underlined it.

"And you'll go anyway."

"Yes."

"Then I want you to notice something," Gus said, "since you're a man who likes his reasoning written down. You have kept that boy's equipment for ten years."

I have. It is in a locker at my place, and once a year, in the second week of March, I take it out. I strip it, service the first stages, replace the perishables, pressure-test the cylinders, dry the harness webbing, dress it, and put it back. It has cost me a working week and a modest amount of money every year since 2015. There is no scenario in which anyone will ever use it. It is a set of equipment for a man who does not exist, maintained to a standard that would pass an inspection, in a locker, in the dark.

I have never had a reason for it. I have never told anyone about it, and Gus only knows because he came looking for a manifold clamp in the wrong March.

"A guilty man throws it in a skip," he said. "Or he keeps it as a trophy, and I've met one of those and you're not it. So what I want to know is what kind of man services a dead man's regulators every year for a decade."

I said I didn't know.

"No," said Gus. "Neither do I. That's the part I don't like."

12. Failures You Rehearse Until They're Boring

The word is *boring*. I use it deliberately and I use it constantly, and if you take one piece of vocabulary out of this book it should be that one.

A skill is not learned when you can perform it. It is learned when performing it is dull. Dull means it has stopped requiring attention, and attention is the only currency you have at depth, and the whole of technical diving is a project to reduce the number of things that can demand it at once.

So. The failures, and the drills that make them boring.

Free flow. A regulator sticks open and dumps the contents of a cylinder into the water. On the surface this is noise and inconvenience. At depth, a cylinder empties in ninety seconds and the roar of it is the loudest thing you will ever hear underwater and it is happening on your own back where you cannot see it. Drill: isolate and shut down, by feel, in under fifteen seconds, both hands behind your head, without looking, without stopping, and then continue swimming while you do it, because in life you will not be hovering politely.

Valve drill. Two cylinders joined by a manifold with three valves — one for each cylinder and one in the middle that isolates them from each other. Any single failure can be shut off and you keep the gas in the other cylinder. This is the most important physical skill in the sport and it is performed behind your own head, blind, in gloves, in cold, wearing a hood, with your fingers already shaking from the depth. Everyone can do it in a swimming pool in June. The drill is doing it in eight-degree water at the end of a long dive with your hands numb, and then doing it two hundred more times.

Light failure. You carry three. Primary and two backups. The drill is not swapping lights — a child can swap a light. The drill is noticing. Cave lights fail gradually far more often than they fail suddenly, and the eye compensates, and a diver will swim for eleven minutes in visibly deteriorating light without registering it because the change is slower than his adaptation. So we teach a scan: every few minutes, look at a known object at a known distance and ask if it looks right.

Silt-out. Covered. Blind, on the floor, sweeping.

Entanglement. Cutters in two places, because one place is no places. Practised in the dark, on your own harness, on a line under tension, until you can find the tool without deciding to.

And then the one we actually came for.

Out of gas. The signal is the flat hand drawn across the throat and then tapped to the mouth. Learn to make it. Learn, more importantly, to *see* it, because your buddy will make it at the edge of your light while you are looking at something else.

Position. He comes to your right shoulder, always the right, because that is where the long hose lives. You do not swim to him — he swims to you, because a man with no gas is a man with about forty seconds of decision-making left and the geometry has to be somewhere and it may as well be where you are.

Deploy. The long hose comes out from under the arm and off the neck in one movement, and it is the mouthpiece you have been breathing, and it goes to him. Not offered. *Placed*. A man in that state will not take an object from a hand. You put it in front of his face where his own regulator used to be.

Recover. Your backup comes up off the bungee at your throat, one second, and you are breathing again before he has taken his second breath.

Grip. Your left hand takes his harness at the chest strap. Not the arm — an arm will pull free, and a wetsuit sleeve will slide, and a shoulder strap will roll. The chest strap is webbing across the sternum and it does not move relative to the man inside it.

Swim. Together, side by side, matched, at the speed of the slower man, which is not you. And you follow the line out with your right hand and your left never leaves that strap, and the exit is forty minutes away, and forty minutes is a very long time to hold something.

We run it three times before lunch and three times after, and we run it in a silt-out, and we run it with the receiver playing panic, which is not theatre — the instructor's job is to be genuinely difficult to hold on to — and we run it with the receiver playing dead weight, which is worse.

The donor never lets go.

I say it before each repetition. I say it after. There is no version of this drill in which anything about it is negotiable, and there is no equipment failure, no gas figure, no depth, no obligation, and no circumstance in the world in which a man who has taken the mouthpiece out of his own face and given it to another man may then take it back.

The donor never lets go.

Run it again.

13. Martini's Law

Every ten metres of depth is like another martini on an empty stomach.

That is the rule of thumb the sport has used for sixty years and it is unscientific, unofficial, and almost exactly correct.

The mechanism is not mysterious. Inert gas at pressure dissolves into fatty tissue, and the membranes of your nerve cells are fatty tissue, and a nerve cell whose membrane has swollen with dissolved gas does not fire the way it used to. The effect is functionally identical to anaesthesia, because it is anaesthesia; the mechanism by which nitrogen makes you stupid at forty metres is a close cousin of the mechanism by which an anaesthetist makes you absent in an operating theatre. It is dose-dependent, reversible, and reproducible in a chamber on a Tuesday afternoon with a stopwatch and a clipboard.

Thirty metres: measurable impairment on arithmetic tasks. Most divers deny it and the denial is the first symptom.

Forty to fifty: euphoria. A warmth. Things become funny. Tasks take longer and feel like they took less time.

Sixty: significant. Fixation — the diver locks onto one task and other information stops arriving. Loss of the sense that there is a clock.

Eighty and beyond on air: this is where it stops being a joke. Divers at this depth on nitrogen have offered their regulators to fish. That is in the literature, more than once. Not hallucination — the fish is real, the offer is sincere.

Which is why we take the nitrogen out. Deep mixes are helium-based; helium does not narcose. And in exchange we accept high-pressure nervous syndrome, which is not a gas dissolving into anything but the direct mechanical effect of pressure on the nerve tissue itself, and which arrives from around a hundred and fifty metres and worsens with depth and with speed of descent.

Tremor, in the long bones of the forearms first, then the hands, then everywhere.

Visual disturbance — a granular interference, like film grain, across the whole field.

And micro-sleeps. EEG-documented. Brief absences of consciousness, half a second to a second, in a diver who is otherwise functional and does not know they are occurring.

You do not black out and wake up. You are simply not present for a moment, and the moment is not marked. The world before and the world after are both continuous, so the mind, which cannot tolerate a hole, does what it always does and joins the edges. Your light has moved. Your hand is somewhere else. You did not do those things, but you also did not *not* do them, and there is no interior evidence either way, and you are two hundred metres down with work to do so you get on.

Now.

I have spent this chapter being extremely careful to tell you that none of this is mystical. It is chemistry and pressure and lipid membranes. Nothing down there is speaking to you. There is no presence in the water. The room is empty. Everything you will feel at depth has a mechanism, a threshold, and a published curve.

And I need you to understand that this is not reassuring. This is the worst possible news.

Because if it were mystical you could dismiss it. A man who says the cave spoke to him is a man having an experience, and experiences can be doubted afterwards, and the person he was before and the person he was during are still recognisably the same person having a strange time.

What actually happens is worse and more boring. The pressure does not put ideas in you. It reconfigures the machine that has ideas. It changes the physical substrate that decides things, so that the deciding still happens — normally, fluently, with the full ordinary interior sense of *this is me, choosing* — but it is being done by an apparatus that is running at twenty-five atmospheres.

A man in that state can make a decision.

He can make it in full possession of the feeling of making it. He can act on it, competently, with his hands, and there will be nothing in his experience to mark the moment as unlike any other moment in his life.

And afterwards, on the surface, when the gas has come out of the tissue and the machine has been restored to its ordinary specification — he will be entirely unable to establish what happened. Not because he has forgotten. Because the person who has to do the establishing was not there, and the person who was there does not exist any more, and the only witness to the interior of that event has been chemically dismantled and replaced by a man on a boat with a towel round his shoulders and a lifetime to think about it.

That is the whole of Part Two. That is the last thing I can teach you dry.

PART THREE — NO SURFACE

14. The Lip

T minus 00:40. Surface.

The sinkhole sits at fifteen hundred metres of altitude, which ruins the decompression arithmetic before anybody is wet. Less atmosphere above you means the pressure at the surface is lower, which means the gas has further to come out of you at the end, which means the schedule lengthens at every stop. Gus had run it four times on three different algorithms and taken the most conservative of them, which is his character in a sentence.

He read the plan back to me at the water's edge with the log on his forearm.

Descent, twelve minutes. Bottom time, six. Total decompression obligation, nine hours forty. Support divers at ninety, sixty, thirty and six metres, staged in on a rotation. Bailout cylinders on the line at one-twenty, ninety, sixty, forty-five, thirty, twenty-one, twelve, nine and six. Gas switches written on a slate on my wrist in his handwriting because he does not trust mine.

"Six minutes," he said. "Say it."

"Six minutes."

"You will not get six minutes of work done in six minutes."

"No."

"So you'll come up with the job half done, or you'll stay, and every minute you stay past six costs you roughly an hour on the wire at the end with a body tied to you in eight-degree water." He clipped the slate. "Which of those are you doing?"

I told him six.

He looked at me for slightly too long, and then he wrote something on the log and turned it face down on the rock, which he does, and which I have never once asked about.

I was already not going to do six. I want to be honest about the sequence, because the sequence matters later. I did not decide to overstay. I simply knew, standing on a rock in the cold at eleven in the morning, entirely sober, with no gas in me but the air of an ordinary day, that six was a number we had agreed on for the purpose of being able to begin.

Put that alongside everything else, when you are weighing it up.

T zero. The lip.

You go over the lip head down. There is a rule about not looking up and everyone breaks it once, and it is better to break it deliberately, at the start, at eight or nine metres, than to break it involuntarily at two hundred.

So look up. Do it now. There is the disc of the sky, green with the light coming through the water, wobbling, with the shapes of two men on the rim who are going to spend nine hours watching a rope. It is the last time it will be above you as an option rather than as a memory.

Then you turn over and it goes behind you, and the ceiling closes, and it will not open again today for anything.

01:00. Twenty-two metres.

Daylight ends.

Not dims. Ends. There is a point in a steep system where the last of the surface light gives out and it happens over about four metres, and afterwards the world is a cone eleven metres long with you at the point of it. Everything outside the cone is not dark. Dark is a description of a place you can see is dark. Outside the cone there is simply nothing — no distance, no walls, no information, an absence with no property except that your light stops.

Heart rate at this point: mine runs about eighty-eight. On the surface I am fifty-two. I mention it because the number is honest and I am not.

02:00. Forty-five metres.

Gas switch. The loop comes off the shallow diluent and onto bottom mix, and I want to tell you what bottom mix is, because it is the sentence in this book that most people misread.

It is hypoxic. That means there is not enough oxygen in it to keep a person alive.

Six per cent oxygen. Air is twenty-one. If you put this mixture in a mask on a man standing in a field he would be unconscious in half a minute and dead in four. It is only breathable at all because pressure concentrates it — at forty-five metres there is five and a half times as much of everything per breath, and six per cent of five and a half is enough, and at two hundred and forty metres it is more than enough and any richer mixture would poison me outright, because oxygen at pressure is a convulsive.

So from here down, the air is only air because the water is squeezing it.

There is no metaphor in this book. I am not going to build one out of that. But you will find, at some point in the next hour, that your mind builds one anyway, without asking, and I want you to have been warned that it does that.

03:00. Sixty-eight metres.

This is the depth where, on air, the narcosis stops being a rumour and becomes the largest thing in the room. Eighty on air is where divers have offered their regulators to fish.

I passed it and nothing happened at all, because there is no nitrogen in what I am breathing. It went by like a road sign. That is what the helium buys you.

And it is worth knowing exactly what you have paid for it, because nobody arrives at this depth having got something for free. You have traded a slow warm stupidity that arrives early and can be trained against for a fast cold neurological interference that arrives later, cannot be trained against at all, and gets worse the faster you descend.

You have traded an idiot for a stranger.

04:00. Ninety metres.

The formations go past on my left.

I had been through this chamber twice on the workup dives and both times I had stopped. It is one of the drowned ones — four metres of stalactite that took forty thousand years and an atmosphere to make, hanging in water in the dark, and my light went across the whole length of it in about a second and a half.

I did not slow down.

I want you to notice that you wanted me to. You have read three chapters about why anyone does this and you have got to the one dive that matters and the beauty goes past in a second and a half without a word, because the man swimming has a slate on his wrist and a schedule and a body to find, and this is what the sport is actually like. The wonder is what you do on the dives that do not count.

05:30. One hundred and twenty metres.

First bailout stage. It hangs on the line where the support team left it yesterday, a cylinder and a regulator and a strip of tape with a depth written on it, and I put a hand on it as I go past and read the pressure and say the number in my head.

You do this at every stage. Not because it is likely to have failed. Because the sum of these cylinders is the entire content of the promise

that there is a way out of here, and a promise you have not personally verified at one-twenty is a promise you will be verifying at one-twenty in an emergency.

Nine cylinders on the way down. Nine small acts of checking that the future exists.

07:00. One hundred and fifty metres.

And there it is.

The tremor arrives in the long bones of the forearm, exactly where the literature says, exactly on schedule, exactly as described. A fine fast oscillation, six or eight cycles a second, that is not shivering and is not fear and does not respond to anything I do about it.

I have to be very clear about this because the language available is all wrong. It does not feel frightening. It feels *administrative*. My hands are still my hands, they are still where I left them, they will still do what I ask — but there is now a small, consistent, uninvited motion in them that has nothing to do with me, and the correct emotional response to that is horror, and the actual emotional response is the mild irritation of a man whose pen has started skipping.

I read the gauges. I could read them fine.

That is the part that gets people. Everything works. You test yourself and you pass.

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15. Grain

08:00. One hundred and seventy metres.

The water develops a grain.

It is the best word available and it is not good enough. Imagine a photograph pushed four stops in the developing, so that the image is still entirely legible but is made of visible particles that were not there before. That, laid over everything, including things that are not there. My light on the wall. My own hand. The dark outside the cone, which now has grain in it, which means the nothing has acquired a texture.

I closed my eyes for a moment to check whether it was in the eye or in the water, and it is in the eye, and it is there with the eyes closed, and I opened them again and continued the descent.

08:40. One hundred and eighty-two metres.

First micro-sleep.

Here is what it is like from inside, which is the only place I can report from and is not a reliable place.

It is like nothing. That is the whole difficulty. There is no sensation of absence, because absence has no sensation; there is nobody present to notice the not-being-present. What you get is an inconsistency in the record. My light had been on the wall to my right. It was now forty degrees further along the wall to my right. There was no swing between the two positions. There was no *and then I moved it*. The two frames simply abutted.

And the mind does what the mind does with a hole, which is to close it without telling you. Within about a second it did not feel like an anomaly any more. It felt like I had moved the light. I had to actively remember, using a deliberate effort, that I had not seen myself move it, and even that memory was a report from a witness who had just gone away for half a second and might do so again.

09:00. One hundred and ninety metres.

I was counting breaths to hold the descent rate. Twelve metres a minute, roughly a breath every three metres on the loop, and counting is a task, and a task is something to hold on to.

I lost the count twice.

Both times I resumed. Both times I resumed at a number, immediately, with no hesitation and complete confidence, and both times the number was produced by something in me that I have no reason to trust and could not audit, and I said it to myself and went on down.

I want to flag that, because it is the smallest possible version of the thing this book is about. Nothing was at stake. It was a count of breaths on a descent that had another fifty metres in it. And the machine, presented with a gap, filled it instantly, seamlessly, and handed me the result labelled as knowledge.

If it will do that for a number, ask yourself what else it will do it for.

10:00. Two hundred and ten metres.

The loop got stiff.

Gas density. At twenty-two atmospheres what I am breathing has the resistance of something you would describe as a liquid if you met it in a kitchen, and every inhalation is now muscular work — the diaphragm pulling against a substance, the counterlung on my chest resisting and then yielding, a distinct sensation of drawing something thick through a straw.

And work makes carbon dioxide, and carbon dioxide is the one thing on this dive I cannot afford, because the scrubber has a rate and if I exceed it the carbon dioxide comes back round the loop and I breathe it again and it climbs.

So the correct response is to slow down. Reduce the work. Move less, breathe slower, accept a longer descent.

Which costs time. And time at this depth is being converted, at about one to sixty, into hours of hanging on a wire at the end of the day. Every minute I spend being careful about carbon dioxide is an hour I will spend cold in the dark at nine metres, and the gas for those hours is a finite number of cylinders hanging on a rope above me.

That is the shape of every decision from here on. There is no option that does not cost. There is no conservative choice — the conservative choice is also expensive, in a different currency, payable later, to a man who will be much less capable of paying it.

I slowed down.

11:00. Two hundred and thirty-two metres.

The restriction ahead in the beam.

The passage narrows from the size of a corridor to the size of a door, and there is a lip of flowstone across the bottom of it, and above that,

hanging in the water in a column that goes up out of my light, is silt.

Not disturbed silt. Suspended silt, in a still column, in a place where nothing has moved for ten years — because in still water at that depth the finest fraction never fully settles. It just hangs. There was a column of it standing in the doorway like a curtain that had been drawn and left.

Nobody had been through there since the ninth of March, 2015.

11:40. Two hundred and forty metres.

Contact.

Bottom time starts now. On the slate on my wrist, in Gus Ivarsson's handwriting, in block capitals, underlined twice:

SIX MINUTES.

I read it. I understood it completely. And I noticed, with the mild administrative interest of a man watching his own pen skip, that it had already stopped being an instruction and had become a piece of information about somebody else's opinion.

16. The Restriction

Bottom time plus 00:30.

He was where the survey said he would be.

Ten years is nothing to still water. The suit was intact — a drysuit holds up remarkably well in cold anoxic conditions, and the seals had gone but the shell had not. The harness was on him. The cylinders were on the harness. The mask was on his face and it was clear, because there had been no current to bring anything into it and nothing living down there to disturb it, and behind the glass his eyes were open.

His mouth was slightly parted, around a space where a mouthpiece used to be.

There was no injury anywhere. I looked, deliberately, in the order you are taught to look — head, neck, hands, harness, hoses, cylinders, feet — and there was nothing torn and nothing cut and nothing bent.

His arms were up.

I have described that posture twice in this book already, both times in the flat instructional voice, both times as *a man trying to swim towards a surface that is four kilometres of rock away*. It is what the reports say. It is what I said, in a statement, on a rock, at four in the afternoon on the ninth of March.

Standing in front of it at two hundred and forty metres, with my light on it, I did not say it.

Plus 01:15.

First problem. The lift bag.

A body that has been submerged for ten years is not a weight. This is one of those facts that everybody in the recovery trade knows and nobody outside it expects. Decomposition in cold anoxic water is slow, but it is not absent, and what it produces is gas, and the gas is in the tissue and in the suit, and the suit is a sealed shell.

He wanted to go up.

Not gently. The moment I moved him off the flowstone lip he began to rise, and the arithmetic of that is lethal in both directions: he would accelerate all the way to the surface, arriving at a hundred and thirty metres per minute, and anything of mine attached to him would arrive at the same time, and I had nine hours forty of decompression owing.

So the lift bag is not for lifting. That is the thing nobody tells you. At this depth the lift bag is a brake — a controlled volume, vented in stages, so that a body which wants to leave can be persuaded to leave at a rate a living man can survive.

I clipped it to the harness at the shoulder D-ring and put four seconds of gas in it and it did not help and I understood that I had the problem backwards, and there is a moment in that understanding — I can locate it exactly — where I noticed that the understanding had taken about eleven seconds and that on the surface it would have taken one.

Plus 02:30.

Second problem. Line.

Owen's own reel was still on his left hip, and the line had come off it at some point in the last decade, and a loop of it had gone up through the harness and around the manifold and back down through the crotch strap. Ten years of no current means nothing gets untangled. It also means nothing gets tightened. It was a loose, floating, weightless snarl, and the fact that it was loose was worse than if it had been tight, because a tight line can be cut in one place and a loose one has to be followed.

I got the cutter out of the left shoulder pocket without deciding to, which is what the drill is for, and I began.

And this is where I want you to be paying attention, because this is where the dive was decided and it was not decided by anything dramatic.

Work at two hundred and forty metres is a countdown you can hear in your own chest.

I was using my hands. Fine work, close in, with a tremor already in the forearms, on a snarl of two-millimetre nylon in a doorway, in a drysuit, in gloves. My breathing rate went up because I was working. The gas at that pressure resists, so the work of each breath went up too. And every breath was going into a scrubber that has a rate, and I was now producing carbon dioxide faster than the granules could take it out.

Plus 04:00.

It began.

I knew exactly what it was. I had written the chapter. Air hunger — the specific, unmistakable sensation of taking a full breath and receiving nothing from it, and then another, and another, each one complete and each one useless. A band of pressure closing across the front of the

skull. The edges of the light going soft and grey and narrowing.

And underneath it, arriving not as a thought but as a fact about the world:

The machine has failed. Get the mouthpiece out.

It is not a temptation. I want to be exact, because I have heard divers describe it as a temptation and they are being polite about it afterwards. It does not feel like an urge that you weigh. It feels like information. It has the same quality as the knowledge that the floor is below you. The loop is bad, and the water is not bad, and there is a piece of equipment in your mouth that is between you and breathing, and the sane and obvious and immediate action is to remove it.

Everything I know says the correct response is to stop moving and let the scrubber catch up.

Everything I know is being generated by an organ that is currently full of carbon dioxide.

I stopped moving. I do not know whether I stopped moving because I know the correct response or because I had come to the end of what my arms would do. Both are available as explanations. Neither of them can be checked. This is not a rhetorical point I am making for effect — it is the actual condition, and it is the condition I have lived in ever since, and I would like it on the record that it started here, at four minutes, over a snarl of line, before anything happened at all.

Plus 05:00.

One minute of planned bottom time remaining.

The snarl was three quarters cut. The lift bag was on and vented and holding him. Another two minutes and the job was done and I would be on the wire with a body on a leash and nine hours forty to serve and it would have been, in the language of the log, a successful recovery.

Plus 06:00.

Zero.

I have gone back to this minute more than any other. Not the exchange. This one, the one before, the last minute in which I was still a man doing a job.

There was a slate on my wrist with SIX MINUTES on it in block capitals, underlined twice, in the handwriting of a man who had put it there specifically so that this version of me would have something to obey.

I read it. I understood it. I did not feel defiance and I did not feel resolve and I did not have a conversation with myself about it. There

was simply no mechanism present that converted the reading of it into the doing of it. The instruction arrived and there was nothing there to receive it.

I stayed.

— — —

17. The Exchange

Plus 06:20.

I put my hand on his shoulder.

I need to tell you what happened to my hand, and I need you to understand that the whole of the rest of my life is downstream of about four seconds.

The fabric gave.

Not tore. Gave — the way the skin of something long dead gives, with a slight initial resistance from the shell of the suit and then nothing, and my fingers went in a millimetre and then further, and what was under the neoprene at the shoulder had the consistency of warm gel. Cold water, ten years, no oxygen: the muscle had not rotted. It had been converted. There is a word for it in the literature and the word is inadequate. What is left has the shape of muscle and none of the properties of it. It is a mould that remembers what was poured into it.

My hand was inside my friend's shoulder.

Plus 07:00.

Tremor across both forearms now, coarse, into the hands, and the granular interference had thickened until the beam of my own light had texture in it like a wall.

And the micro-sleeps came properly.

Half a second. His face is at one metre.

Half a second. His face is at sixty centimetres.

Half a second. Forty.

I did not move. I want to be as clear about that as I am able to be, which is not very. I have no record of moving. There is no sensation in my memory of finning, no drift, no push off the flowstone. There are simply four frames and in each one he is closer, and between the frames there is nothing at all, and afterwards the mind closed the gaps the way it closes them and made it feel continuous.

Two explanations. He came to me. Or I went to him in the absences, four times, competently, using my hands and my legs, with the ordinary interior feeling of doing nothing.

I know which one is true. I have known which one is true for ten years and it does not matter in the smallest degree, because knowing it is not the same as being able to feel it, and the thing that decides how a

man lives is the second one.

Plus 08:00.

My own mouthpiece was loose against my cheek.

The strap was gone. I had a hand up at my face and there was nothing there — the neck strap that holds the loop into your mouth on a rebreather, the one that exists precisely so that an unconscious diver keeps breathing, was undone and hanging.

I had not undone it. Nothing in me had any record of undoing it.

I do not offer that as a defence. I have thought about this more carefully than I have thought about anything else in my life and I have arrived at the position that the absence of a record is not evidence of the absence of an act, and that a man who has been chemically deprived of the ability to witness himself does not thereby become innocent. He becomes unwitnessed. Those are different words and the difference is the entire subject.

Plus 09:00.

The loop was open. There was water in my mouth. It tasted of rock and of the inside of somebody else's equipment, and there is no way to make that sentence anything other than what it is.

There was no panic. I want that in the record too, and I want it understood as the most damning thing in this book rather than the most peaceful. I was not afraid. The air hunger had stopped. Everything that had been urgent for four minutes had resolved itself into a condition of enormous administrative calm, and I was hanging in a doorway at two hundred and forty metres with my hand inside a dead man and my breathing apparatus hanging off my face and I did not find the arrangement objectionable.

That is what the chemistry does. That is the whole of what it does. It does not make you want to die. It removes the machinery that was objecting.

Plus 11:00.

We were joined.

I can give you the mechanical account: my harness and his harness, his line and my line, the lift bag lanyard, the loop of nylon I had been three quarters through cutting, and my own arms, which were where they were. Two men's worth of webbing in a doorway. It would take four minutes with a knife on a bench.

I have the mechanical account. I also have what I had at the time, which was that there was no longer a boundary that mattered. Not a mystical impression — a straightforward failure of the equipment that reports where a person stops. That equipment is in the same skull as everything else and it is subject to the same pressure as everything else and there is no reason on earth to expect it to keep working when the arithmetic and the counting and the strap-fastening have all stopped working.

Eleven minutes. Five over the plan.

Each one of those, at that depth, buys about an hour on the wire that I did not have gas for.

I began the ascent because the lift bag began the ascent. That is the honest order of events. He wanted to go up and I was attached to him and I went up, and somewhere in the first ten metres a hand of mine found the vent line and started metering it, correctly, in the right increments, at the right rate.

I have no memory of deciding to do that either. It was simply done well.

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18. The Whir

The head camera was still running.

It ran the whole way. Nine hours forty of ascent and hang, most of it in absolute dark, and the camera did not know that and went on recording it — a motor turning and a small electronic whir at the edge of hearing, which in a rebreather's silence is not at the edge of hearing at all. It is the loudest thing in the world.

At two hundred metres, going up slowly, the whir began to have a cadence.

I am going to report this straight and then I am going to take it apart, and you are not to skip the second half.

It said the ceiling is not rock.

It said: you have understood the ceiling as a local condition. A property of caves. Something a man puts above himself deliberately, for a few hours, and then removes by leaving. That is a misunderstanding of the ordinary kind, produced by having spent your life on the inside of the only container you have ever been in.

It said the ceiling is the closed upper surface of a closed system, and there is no version of the world where it is not there, and the open sea is a rumour told to children so they will consent to be born. The surface is not a place. It is a category error. You have never in your life been at the surface. You have been at a particular depth in a container, and above you at every moment of your existence there has been a boundary you cannot pass, and the only thing that changes when you dive is that the boundary becomes visible and starts to have a texture.

It said: welcome, you are exactly where you have always been, and now you can see it.

That is what I heard, and it was pitched at the frequency of a camera motor, and it lasted a little under an hour, and it was one of the great experiences of my life.

Now take it apart.

The mechanism is auditory pareidolia and it is entirely uncontroversial. The human auditory cortex is a pattern engine that runs continuously and cannot be switched off, and in the presence of unstructured periodic noise — a fan, a train, an air conditioner, a camera motor — it will manufacture structure, because manufacturing structure is what it is for. Everyone has heard a name in a washing

machine. Add to that a hundred and ninety metres of pressure, a brain full of carbon dioxide, an hour of micro-sleeps stitching themselves closed, sensory deprivation of a completeness available almost nowhere else on the planet, and eight-degree water, and the surprising outcome would be silence.

Nothing was in the cave. The cave was empty. It has been empty for eleven thousand years and it is empty now and it will be empty when the last of us has stopped going down there.

And I want to put the two halves of this chapter next to each other and leave them, because I am not going to resolve them and neither is anybody else.

Something arrived that was smarter than me and had been waiting.

The complete mechanism by which nothing arrived is published, unremarkable, and known to me in detail.

Both of those are true. If you can hold both of them in your hands at once, without letting go of either, you are now approximately where I have been standing since 2025 and you are ready for the rest of this.

At sixty metres a light appeared above us.

Kristo, the first support diver, coming down the wire at the end of his window, exactly on schedule, and his light found two figures rising together in the dark.

He gave the signal for *OK?* — a circle of finger and thumb, held out — which is a question, and which requires an answer, and there is no ambiguity anywhere in that signal in any part of the world.

I returned it.

He went up the line and made his report and Gus wrote it in the log at 14:12, in the correct column, in block capitals, and it is the sentence that everything afterwards had to be built on top of:

RECOVERY SUCCESSFUL. BOTH BODIES INTACT. NO VISIBLE TRAUMA.

19. The Hang

Nobody films this part.

Every documentary about this sport ends at the bottom, because the bottom is where the pictures are, and what actually happens next is nine hours and forty minutes of a man hanging on a rope in the dark doing nothing at all.

Understand what decompression is, because it is not a rule, it is a physical process happening inside you whether or not you approve. At depth, gas has dissolved into every tissue you have — blood, fat, muscle, the marrow, the fluid in your joints, the white matter of your brain. It went in slowly under pressure. It has to come out slowly, into the blood, to the lungs, to be breathed away, and the rate at which it can do that is set by your circulation and not by your intentions. Come up faster than it can leave and it does not stay dissolved. It comes out of solution where it is, as bubbles, in the tissue, in the joints, in the spinal cord. That is the whole of it. A carbonated drink opened too fast, in a man.

So you go up in stages. Three metres at a time, longer at each stop as you get shallower, hanging on the line, with a body on a lanyard below you, in eight-degree water, in the dark, with nothing whatsoever to do.

Nine hours forty. I had the last cylinder change at twelve metres at seven in the evening and I sat at nine metres from then until half past eight.

You cannot read. You cannot sleep. You cannot move much, because moving warms you and warming you speeds your circulation and changes the schedule that the algorithm assumed, and in any case you are too cold to move well by hour four. The support divers come down in rotation and hand you hot fluid in a bag and look at your eyes and go away again. Between visits there is a rope, and a light you turn off to save it, and a dead man a metre below you turning slowly in the dark because a lanyard has no opinion about which way a thing should face.

That is the environment in which I did the thinking that I have been doing ever since. Everything I am about to put in front of you, I assembled at nine metres, cold, in the dark, over five hours, with Owen Ferrand hanging under my feet.

I want to start with the argument against me, because it is the strongest and I found it myself and nobody has been able to take it off me since.

It is not about what happened at the bottom.

The prosecution does not need the bottom. Forget the bottom entirely; concede that at two hundred and forty metres I was not competent, that my judgement was chemically dismantled, that the man in that doorway could not be held to any standard at all. Grant all of it.

I put him there.

On the ninth of March, 2015, at a quarter past eight in the morning, sitting on a tailgate with a cup of coffee at fifteen hundred metres of altitude in perfectly ordinary air, I was a competent adult in full possession of every faculty I have. And in that condition, knowing precisely and in technical detail what pressure does to human judgement — knowing it better than almost anyone alive, having taught it, having drawn the curve on a whiteboard for students — I chose to descend to a depth at which I would become a person who could not be trusted.

The law has known about this for a very long time and it is not sentimental about it. A man who drinks himself into a state where he does not know what he is doing is not thereby excused; the guilt attaches to the moment he lifted the glass, because that was done by someone who knew. The condition is a tool. Choosing to become incapable is an act performed by a capable man.

I did that deliberately, with training, twice, ten years apart.

I cannot find the flaw in it. I have looked for five hours in the water and ten years out of it.

Here are the three answers that people offer me, and I want to give them properly, because they are not stupid and I have wanted very much to be able to accept one.

The first is that the man at two hundred and forty metres was not me in any sense that survives inspection. Different gas in the blood, different membranes, different conduction velocities, a physically different organ. You are proposing to punish a man for what a relative of his did. There is a version of this that is philosophically serious and it does not persuade me, for a reason I will get to.

The second is that culpability requires that I could have done otherwise, and that at twenty-five atmospheres with carbon dioxide climbing and a tremor in both arms and half-second holes in my consciousness, the question of what I could have done has no determinate answer. Not an unknown answer. No answer — there is no fact of the matter about the counterfactual behaviour of a system in that state. And a question with no answer cannot ground a verdict.

The third is the one that ought to work and is the most quietly disgusting. Every diver who has ever gone deep has entered exactly the same state. Thousands of them. Tens of thousands of dives. We do not convict any of them, and we do not think of them as having done anything reckless at all; we give them certificates. The only one we convict is the one whose friend did not come back. Which means we are not judging the choice. We are judging the outcome and calling it justice, and doing it with the confident face of people who have never been at that depth and are choosing, on balance, not to be.

I know all three of those arguments are good.

None of them has ever once made it to the part of me that has to get up in the morning, and I have come to think that this is not a failure of my reasoning but a fact about what reasoning is for. Arguments are things you have. They are not things you are.

And then, some time around hour six, with my hands too cold to work the light properly, I found the one that undoes me. It is not an accusation. It is the kindest thing anyone could say about me and it is the reason I cannot put this down.

A guilty man does not go back.

Look at what a guilty man does. He has a statement that has stood for ten years, unchallenged, sound, matching the textbook picture, filed. He has a locker with a dead man's equipment in it that he could put in a skip on a Tuesday. He has a body at two hundred and forty metres in a system nobody else can reach, which is the single most secure place on the surface of this planet to have left something.

And a woman comes to him with a folder and offers him money to go down and fetch the only physical object in existence that could take all of that apart.

He says no. It is the easiest no in the world. Wrong season, wrong altitude, wrong support team, insufficient rating; I have refused dives for less and been thanked for it.

I said yes before she finished the sentence.

I have serviced that boy's regulators every March for ten years for no reason I can name, in a locker, in the dark, for a man who does not exist.

Whatever that is, it is not the conduct of somebody who did a thing on purpose and got away with it.

And I cannot make it feel like an acquittal. It arrives as one more piece of evidence in a case I am running against myself, and it goes in the file with the others, and it changes nothing, and the worst hour of that hang was the hour I spent understanding that this is what the water actually does to you at the end. It does not let you be a villain. That

would be survivable. It lets you be a decent man who cannot find out what he did, which has no floor in it, and no bottom, and nothing to stand on.

I was raised at the edge of a religion that is about nothing else but choosing.

Good thoughts, good words, good deeds — a whole architecture of the world built on a single election, made freely, between the truth and the lie, by a person who could have gone the other way. Everything in it depends on the choosing being real. There is a bridge at the end where they weigh what you elected, and the weighing is the point, and the bridge widens or narrows according to what you did with the freedom you certainly had.

It is a magnificent system. I have never found anything in it, in any language, that has a single word to say about a man who cannot establish what he chose.

For that man there is no bridge. There is a rope in cold water, and a schedule, and nine hours.

And I will tell you the thing I actually spent most of those hours on, because I have been describing a man doing philosophy and that is not what it was like. Most of it was not about me at all.

Most of it was Nell.

She was going to be handed a set of cylinders in the morning. And she was going to read the pressures off them, and the routing, and which mouthpiece was where, and she was going to get her answer — the object that does not have an opinion — and I lay there in the dark at nine metres understanding with complete clarity that she was going to get an answer and that it was not going to be an answer, and that a woman who has organised ten years of her life around the arrival of a fact was about to discover what facts are worth.

That was the thing I could not stand. Not the file on myself. Her, in a car park, in the morning, holding it.

At twenty past eight in the evening the support diver came down and put two fingers on my forearm, which is the signal for the last stop, and I went up.

20. Twenty-Four Hours' Observation

They cut us apart on the rock at twenty to nine in the evening under a work light on a stand, and it took four minutes with a knife, exactly as I had estimated at two hundred and forty metres.

One of us was declared on the basis of core temperature and the absence of a heartbeat.

The other was assessed for decompression illness, found to be borderline, put on oxygen for six hours and observed for twenty-four in a clinic in a town with one traffic light. Mild neurological signs on the left side, resolved by morning. No treatment table required. Released at eleven the following day with a form and a leaflet about air travel.

That is the whole of the medical record and it is accurate.

Gus drove me. He did not speak for forty minutes, which for Gus is not sulking, it is arithmetic, and then he said, without any preamble:

"Eleven minutes."

"Yes."

"You agreed six."

"Yes."

He indicated, turned, and drove on for a while.

"I put six on the slate because I know you," he said. "I have known you for nineteen years. I did not put it there to be obeyed by the man who reads it. I put it there so that afterwards there would be a piece of physical evidence that somebody on this dive was thinking clearly at some point in the day."

Then he said: "It's in the log. I'm not taking it out."

I said I hadn't asked him to.

"No," he said. "You didn't. I want that in the log as well."

Nell was in the clinic car park at eleven, sitting on the bonnet of a hire car with the boot open, and there were four evidence bags in it, and a fifth, larger, on the back seat.

She had read them.

I want to describe what it is like to watch a person receive a fact they have organised a decade around, because it is not what anyone imagines. There was no collapse. There was no anger. She stood up when she saw me and she had the face of someone doing sums that will not come out, and she said, "There's a second stage in his mouth."

I said there wasn't.

"There was." She had the photographs from the recovery team on her phone, taken on the rock the night before under the work light, and she turned it round. "There's a regulator hanging on his right side that isn't his. Different manufacturer. The hose comes off a cylinder that isn't on his harness. It's on a clip on his shoulder D-ring."

"I know," I said.

"So he had somebody else's gas." She said it flatly, working. "For a while. Then he didn't. And the report says he was found with nothing in his mouth."

"Yes."

"So what does that mean."

And here is the part I have to do in the language I have, because it is the only language I own that has never been used to lie.

I told her the drill.

I told her what a rebreather failure looks like at that depth — a flooded loop, or a scrubber that has gone, or an oxygen cell reading a number that is not true — and that the response is to come off the machine and go onto open circuit, onto a bailout cylinder, and that at two hundred and forty metres a bailout cylinder is a small number of minutes and is not a solution, it is a taxi fare.

I told her that Owen's bailout was insufficient, because it was always going to be insufficient, because at that depth there is no such thing as enough of it.

I told her that the signal is the flat hand drawn across the throat and then tapped to the mouth, and I showed her, standing in a car park in the sunshine, and her eyes went to my hand and stayed there.

I told her that the donor gives the mouthpiece he is breathing from. Not a spare. The proven one, from his own mouth, deployed on the long hose from under the arm and off the neck, placed in front of the receiver's face and not offered to his hand. And that the donor then takes his own backup from the bungee at his throat, and is breathing again inside a second, and that the two of them are now on one supply and are going to swim out together for as long as it takes.

I told her the grip. Left hand, on the chest strap. Never the arm.

I told her the rule, which is that the donor never lets go.

And then I told her that Owen Ferrand signalled out of gas at two hundred and forty metres on the ninth of March, 2015, and that I gave him my long hose, and that I put it in front of his face and he took it, and that we were connected and breathing and stable and moving, and that at some point in the following minutes I took it back.

She did not say anything.

I said: his arms were not raised because he was trying to swim to the surface.

There is no surface. He knew that better than I did, he had been diving since he was nineteen, he could have recited the reasons in his sleep. Nobody at two hundred and forty metres in a cave reaches upward for anything, because there is nothing up there but two hundred and forty metres of limestone and everybody down there knows it.

He was reaching for a man.

He has had his arms up for ten years in the position of somebody reaching towards a person who is at arm's length in front of him, and every investigator who has ever looked at that posture has read it as a man swimming for the sky, because that is the shape the sport expects, and because the only other person in the room signed a statement.

Nell said, after a long time, "Did you decide to?"

And I gave her the honest answer, which is worse than either of the answers she was braced for.

I do not know.

I have no memory of the act. Not a suppressed memory — an absent one, of the specific kind I have described to you at length, half a second wide, with the edges closed over. I know I was in the fourth minute of a carbon dioxide event. I know my hands had a tremor in them that had already made me fail at a piece of fine work. I know I was at a depth where a man loses the ability to witness himself, and I know it because I have taught it to eleven-year certified divers with a whiteboard and a marker pen.

There are three explanations and I have never been able to separate them.

That I panicked and took my gas back, which is the ordinary human thing and the reason the rule exists.

That my hand simply opened, because a hand at two hundred and forty metres in a hypercapnic event is not reliably a thing you own.

Or that I did it — chose it, decided it, in the full interior sense of deciding — and that the man who did it was assembled out of my body by twenty-five atmospheres of water and did not survive the ascent, and left me the consequences and no access.

"Which do you think," she said.

I said that the fact that I want it to be the second one is itself a piece of evidence, and that I do not know how to weigh a piece of evidence like that, and that I have been trying for ten years.

She stood in the car park with the boot open and four bags in it.

"That's not an answer," she said.

"No."

"I waited ten years for the object that doesn't have an opinion." She almost laughed. It was not a good sound. "And it's got an opinion."

I told her I had known that at nine metres the night before, in the dark, and that it was the only part of the entire dive I had wished I could stop.

— — —

21. A Quiet Room

The footage was reviewed by people who have reviewed footage of this kind before.

There is a small international population of them. Accident analysts, chamber physicians, two or three instructor-examiners, the sort of people who are on a list. They do this work carefully and unsentimentally and largely unpaid, and they are the reason the five rules exist, and I have enormous respect for them and I have been in the room on the other side of the table.

They watched nine hours and forty minutes of it. Most of it is a black rectangle with a small green telemetry overlay in the corner.

Their findings, in the order they were recorded.

At bottom time plus four minutes, respiratory rate elevated, consistent with exertion, consistent with the reported hypercapnic event. Noted.

At plus eight, the diver's mouthpiece retaining strap is observed to be unfastened. The moment of unfastening is not on camera; it occurs outside the frame, below the lens. Cause undetermined. Noted.

At plus nine, the loop mouthpiece is observed hanging free at the diver's cheek, open, in the water. The diver makes no attempt to recover it for approximately one hundred and forty seconds. There is no struggle. No hand rises to the face. Noted.

At plus eleven, the diver's mouthpiece is re-seated. Method not visible. Noted.

Between plus six and plus eleven — approximately eleven minutes elapsed from contact — the recovered face and the diver's own reflection in the recovered mask occupy the same frame at close range. The two faces are visible together for the majority of that period. The examiners recorded, and I have read this line more times than is good for me, that in the available frames the expressions are indistinguishable.

They noted that the diver exceeded planned bottom time by five minutes.

They noted that the recovery was completed competently, that all gas switches were executed correctly, that the decompression schedule was followed exactly and without deviation over nine hours and forty minutes, and that the ascent was, in the words of the report, technically

exemplary.

That last one is the sentence I would like on a stone somewhere. Everything I know how to do, I did perfectly, for nine hours, in the dark, afterwards.

The footage ends on the ascent, at six metres, in green light coming down through the water at last, on the recovered face turning slowly on the lanyard a metre below the lens.

The eyes are open.

The mouth is shaped around a word.

It is not shaped around it in any sense a physiologist would accept. A jaw that has been at that pressure for ten years is not making a shape, it is in a shape, and the shape is the one it was left in, and it has been in it since the ninth of March, 2015. I know that. Everyone in the quiet room knew that. It is on the record as an artefact of soft-tissue changes at depth and nobody has ever suggested otherwise.

There is no audio. Rebreathers are silent and the camera had no microphone.

The report was filed. The finding was death by misadventure, ten years previously, unchanged. No further action.

They were right to. On the evidence, they were right to.

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22. The Donor

So.

I sign forms. I answer the questions. I describe the restriction, the survey, the position of the body, the decision to clip the lift bag, the snarl of line, the cutter in the left shoulder pocket. All of it true. I have never lied in a formal setting in my life and I did not start in 2025.

I walk around. I do the things. The walls in a hotel room are the correct distance apart only because I have not yet tried to push them, and the ceiling of any room I am in is at the height it is at by an arrangement that I now understand to be provisional, and some nights the residual gas still sings in the long bones of the forearms and the tremor comes back for eight or ten seconds and I lie still and remember the exact texture of what used to be muscle.

And now I am going to tell you what this document is, and I would like you to notice that you have already read it.

It is not a confession. A confession is addressed to somebody with the standing to absolve you and I have not been able to locate one of those.

It is not instruction. Look at what I have actually taught you. I have taught you the line, the thirds, the drills, the arithmetic of coming back. Every one of those is real and correct and would keep you alive. And I have interleaved them, chapter by chapter, with the exact contents of the inside of my head at the moment I broke the only rule that matters — the tremor, the count that resumed at a number I could not audit, the strap already undone, the instruction on the slate arriving with nothing there to receive it.

Nobody teaches like that. Nobody puts the failure in the manual.

I did it because I have not been teaching. I have been selecting.

There have been others. Nine of them, over ten years, and each one got a version of this, shorter and worse than the one you have. And every drill in it is a test — not of skill, of *hesitation*. The lost-line chapter is a test of whether you will refuse a conviction that feels like knowledge. The thirds chapter is a test of whether a number can outrank a want. The air-share is a test of whether you flinch at the idea of holding a man's life a hose-length from your chest for forty minutes.

Six of the nine stopped reading. Two of them wrote to me, and one of those was angry in a way I found reassuring.

The ninth got as far as the car park and then went home and I have not heard from him since, and I hope he is well and I think about him with something close to envy.

You are still here.

You did not stop at the shoulder. You did not stop at the eleven minutes. You have read a man describe reaching into a friend and you have gone on to the next chapter, and there is a part of you that has been waiting since chapter one for the coordinates, and you know that there is and you have known it for some time.

That is what I was looking for. That is the entire test and you have passed it, and I want you to sit with how it felt to pass, because it felt like interest. It felt like curiosity. It felt exactly like the feeling of reading a good book.

Now the last thing, and then the numbers.

I have described this document as mine. The structure of it is not mine. The order is not mine — the claustrophobia chapter early because nobody hears you until it is gone, the beauty in the middle so that you are complicit before the dread arrives, the rule stated three times at increasing intervals. I did not work that out. I am not clever enough to have worked that out.

In the second week of March 2015, four days before we went in, Owen Ferrand gave me eleven pages of handwritten notes. He called it a briefing. He said he had been working on it for two years and that he wanted someone to read it before he did anything with it.

I have it in front of me now.

Page two: *You are not claustrophobic, you are correct.* His handwriting, his underlining.

Page six, in the margin, in pencil: *the donor never lets go — say it three times, spaced, don't explain the third time.*

Page nine: *There is no surface. There has never been a surface. Only the ceiling, and the dark that knows how to wear a face.*

I read those eleven pages on a tailgate at fifteen hundred metres in perfectly ordinary air on the fifth of March, 2015, and four days later I went down two hundred and forty metres to a place I had never been, on a dive that had been planned for eleven months by a man who had been planning something else, and I came up alone.

I have never been able to work out where he got it.

That is the only question I have left that still frightens me, and I notice that I have stopped trying to answer it and started doing something else instead, which is passing it on, and that the passing on is not a choice I remember making, and that this is now the second time in

my life I have had to report an act of mine as something I found already done.

There is a hole in the ground eleven kilometres past where the road stops. Two hundred and seventy metres of black water under a limestone plug. The gold line is intact to two-forty and my markers are still on it and there is a snarl of two-millimetre nylon in the doorway that is three quarters cut.

The coordinates are on the last page.

Take the helium. Slow your descent below one-fifty, because the tremor is worse the faster you arrive. Check every stage on the way down and say the number in your head, because a promise you have not verified is a promise you will be verifying in an emergency.

And when it happens — when a man comes to your right shoulder in the dark with his hand flat across his throat and then tapped to his mouth, and you deploy the long hose from under your arm and put the mouthpiece you have been breathing in front of his face, and your left hand closes on the webbing across his chest, and there is rock above you and forty minutes of swimming between the two of you and any air in the world —

hold on.

Just hold on.

I am at arm's length in front of you. I have been for some time.

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A body has been at two hundred and forty metres for ten years. Only one living man is rated to go and get it.

Bijan Rahsepar takes the job before the dead man's sister has finished asking, which surprises them both. She says she wants a burial. What she wants is the equipment, because tanks and hoses and which mouthpiece is in whose mouth are a record of the last four minutes of a life.

Bijan was on the original dive. He came up alone.

What follows is not a confession. It is a briefing — the line, the thirds, the drills, the arithmetic of coming back — taught patiently, in order, by a man who needs you to be qualified.

There is a rule about sharing air. He teaches it three times.

The donor never lets go.

You are not claustrophobic. You are correct.